# Biomagnification: A Sweets Model

*See how a poison becomes more concentrated as it moves up a food chain*

**AIM**
To model how a pesticide such as DDT builds up at higher levels of a food chain, using sweets and cups.

## Materials (per team)

- 1 packet of **Gems** (or similar small sweets, beads or dried beans): 30 pieces, of which exactly **10 are one colour** (say red). *Red = DDT-contaminated.*
- **1 large shaker cup** — the producers (e.g. grass)
- **9 small cups** — grasshoppers
- **3 medium cups** — frogs
- **1 large cup** — the top predator (snake or osprey)
- data table (below), pencil, calculator

**TEACHER PREPARATION**
Count out the 30 sweets for each team beforehand with exactly 10 red. Teams must restore exactly 10 red among 30 before every trial. Do not eat sweets used in the experiment; give clean ones as a reward afterwards.

## Procedure

1. **Producers.** Put all 30 sweets in the shaker cup. This is the producer population. Each producer carries on average 10 ÷ 30 = **⅓ unit of DDT**. Write this in the data table.
2. **Mix.** Shake the cup well.
3. **Grasshoppers eat producers.** Without looking, take **3 sweets** from the shaker cup and put them in a small cup. Repeat for all nine small cups (27 sweets eaten, 3 left in the shaker). Count the **red** sweets in each small cup — this is the DDT in that grasshopper. Record it.
4. **Frogs eat grasshoppers.** Each frog eats **3 grasshoppers**: empty the contents of 3 small cups into one medium cup. Repeat for the other two medium cups. Count and record the red sweets in each frog.
5. **The snake eats frogs.** The snake eats all **3 frogs**: empty all three medium cups into the large cup. Count and record the red sweets in the snake.
6. **Repeat.** Put everything back in the shaker cup, check that there are exactly 10 red among the 30, and repeat steps 2–5 two more times, for **3 trials** in total.
7. **Calculate** the average amount of DDT for each organism type over the 3 trials, and present the result as a bar graph.

# Data table

| Organism | Trial 1 | Trial 2 | Trial 3 | Average DDT per organism |
|---|---|---|---|---|
| **Producer (each sweet)** | ⅓ | ⅓ | ⅓ | ⅓ |
| **Grasshopper (average of the 9 cups)** | | | | |
| **Frog (average of the 3 cups)** | | | | |
| **Top predator (the large cup)** | | | | |

**WHAT RESULTS TO EXPECT**
Each grasshopper eats 3 sweets, so it has about 1 red sweet (⅓ × 3). Each frog eats 3 grasshoppers, so about 3. The top predator eats 3 frogs, so about 9. The numbers climb roughly 1 → 3 → 9. Individual cups will differ because the choice is random (that is why we repeat the trials), but the average should show the trend.

# Discussion

1. What happened to the amount of DDT in each organism as you went up the chain?
2. Why did the poison become more concentrated at each level? (Each predator eats many prey, and the DDT is not removed.)
3. In this model, what assumption did we make about what the animals do with the DDT? How is that like (or unlike) real DDT?
4. Which animal would be harmed first by DDT in the water? Why?
5. How could a farmer reduce the risk? (See *Biopesticides* in the Eco-Lies document.)
6. Why did we repeat the activity 3 times?

## Linking the model to the real world

| In the model | In nature |
|---|---|
| Red sweets | A persistent poison such as DDT, PCBs or methylmercury |
| Shaker cup | Plants and algae that take the poison up from soil and water |
| No red sweet ever leaves | The poison is not broken down or excreted — it is stored in body fat |
| Predator takes in all the prey's sweets | A predator eats many prey and stores their combined poison |

**WHERE THE MODEL IS TOO SIMPLE**
In real life animals have different body masses, and scientists measure the concentration (amount per kilogram of body mass), not just the amount. Real animals also eat more than 3 prey, and some poisons are partly lost from the body. The model shows the idea, not exact numbers.

## Extension ideas

- Change the number of sweets eaten (2 instead of 3) and predict the result first.
- Add a "decomposer" cup that removes half of the red sweets from the environment each round. What happens to the top predator?
- Look up the mercury or DDT story for a local river, lake or fishery.